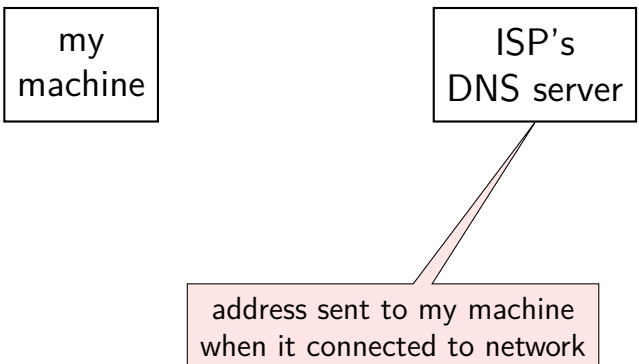
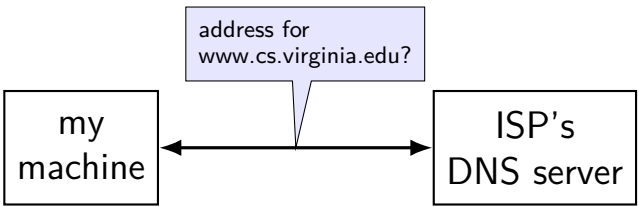
names and addresses

name	address
logical identifier	location/how to locate
DNS name `www.virginia.edu`	IPv4 address `128.143.22.36`
DNS name `mail.google.com`	IPv4 address `216.58.217.69`
DNS name `mail.google.com`	IPv6 address `2607:f8b0:4004:80b::2005`
DNS name `reiss-t3620.cs.virginia.edu`	IPv4 address `128.143.67.91`

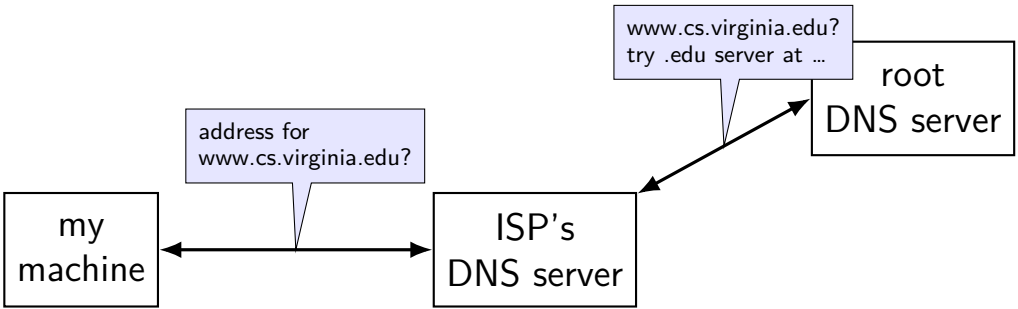
DNS: distributed database



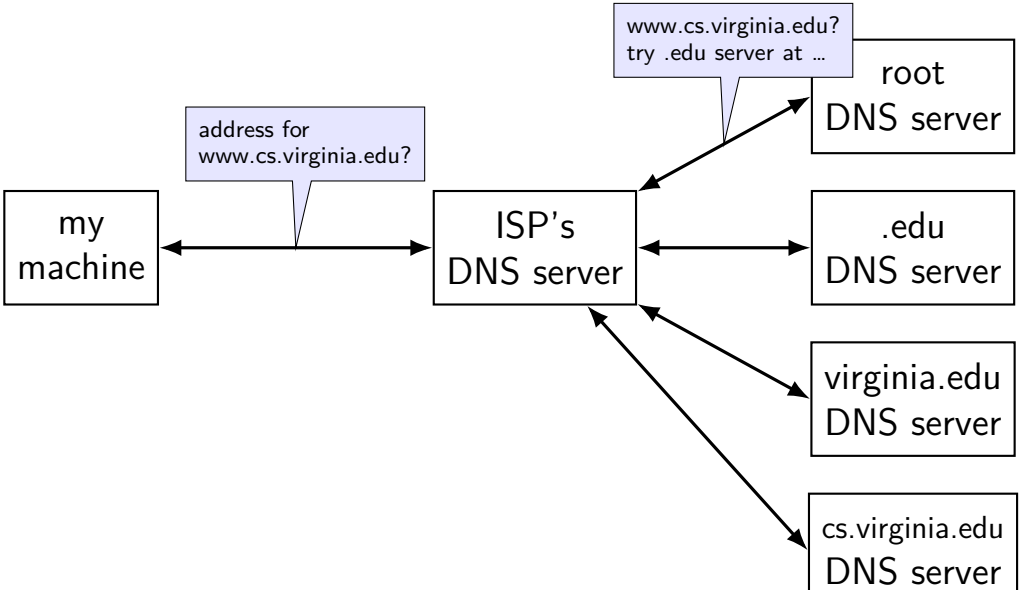
DNS: distributed database



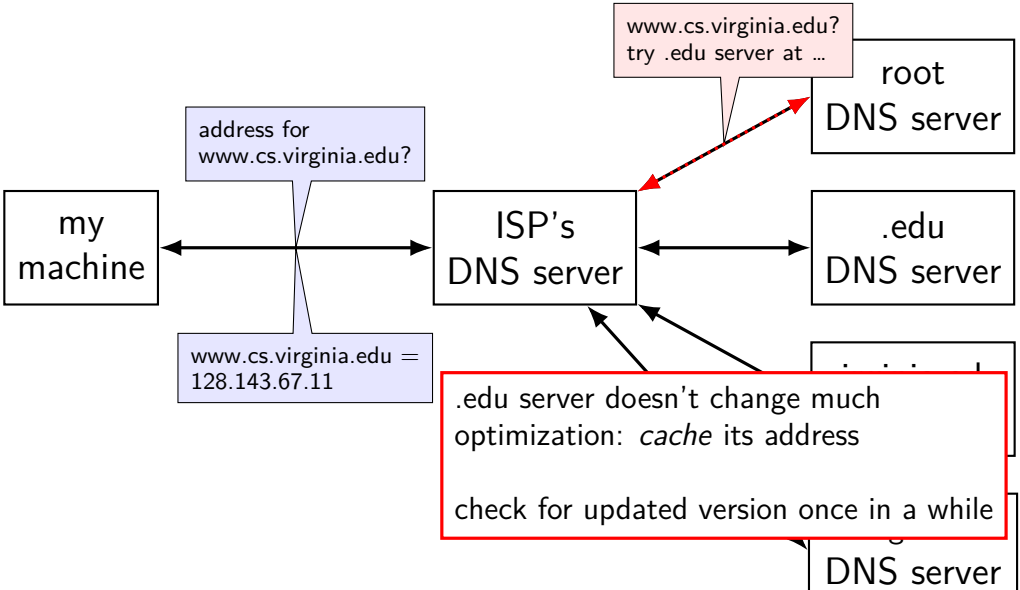
DNS: distributed database



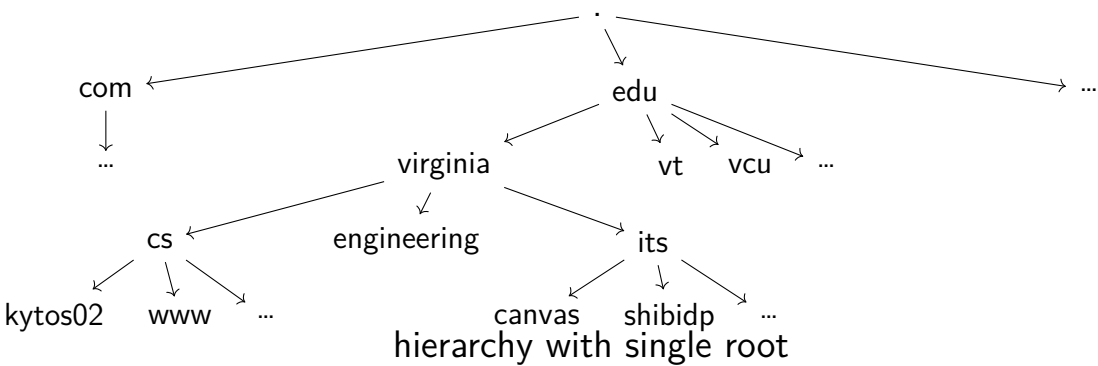
DNS: distributed database



DNS: distributed database

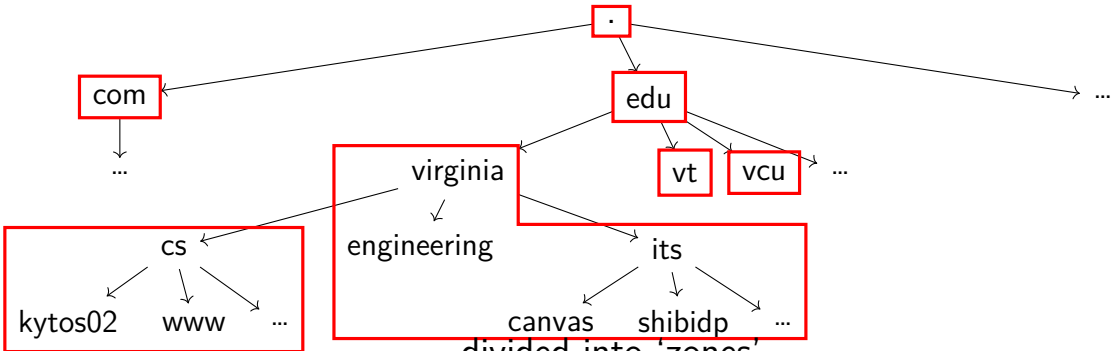


DNS hierarchy



hierarchy with single root

DNS hierarchy



divided into ‘zones’

each with own set of authoritative servers

terms: authority, recursive

DNS server is *authoritative* for X.Y.Z?

(claims to be) official source of information for X.Y.Z

not giving a cached version obtained from elsewhere

DNS server is a *recursive resolver*?

will contact other DNS servers to get answer

choice of whoever configures DNS server

querying the root

```
$ dig +trace +all www.cs.virginia.edu
...
edu.                    172800      IN      NS      b.edu-servers.net.
edu.                    172800      IN      NS      f.edu-servers.net.
edu.                    172800      IN      NS      i.edu-servers.net.
edu.                    172800      IN      NS      a.edu-servers.net.
...
b.edu-servers.net.      172800      IN      A       191.33.14.30
b.edu-servers.net.      172800      IN      AAAA       2001:503:231d::2:30
f.edu-servers.net.      172800      IN      A       192.35.51.30
f.edu-servers.net.      172800      IN      AAAA       2001:503:d414::30
...
;; Received 843 bytes from 198.97.190.53#53(h.root-servers.net) in 8 ms
...
```

querying the edu

```
$ dig +trace +all www.cs.virginia.edu
...
virginia.edu.                 172800        IN        NS        nom.virginia.edu.
virginia.edu.                 172800        IN        NS        uvaarpa.virginia.edu.
virginia.edu.                 172800        IN        NS        eip-01-aws.net.virginia.edu.
nom.virginia.edu.        172800        IN        A        128.143.107.101
uvaarpa.virginia.edu.         172800        IN        A        128.143.107.117
eip-01-aws.net.virginia.edu. 172800 IN        A        44.234.207.10
;; Received 165 bytes from 192.26.92.30#53(c.edu-servers.net) in 40 ms
...
```

querying virginia.edu+cs.virginia.edu

```
$ dig +trace +all www.cs.virginia.edu
...
cs.virginia.edu.        3600        IN        NS        coresrv01.cs.virginia.edu.
coresrv01.cs.virginia.edu. 3600        IN        A        128.143.67.11
;; Received 116 bytes from 44.234.207.10#53(eip-01-aws.net.virginia.edu) in 72 ms

www.cs.Virginia.EDU.        172800        IN        A        128.143.67.11
cs.Virginia.EDU.        172800        IN        NS        coresrv01.cs.Virginia.EDU.
coresrv01.cs.Virginia.EDU. 172800 IN        A        128.143.67.11
;; Received 151 bytes from 128.143.67.11#53(coresrv01.cs.virginia.edu) in 4 ms
```

querying typical ISP's resolver

```
$ dig www.cs.virginia.edu
...
;; ANSWER SECTION:
www.cs.Virginia.EDU.          7183        IN        A        128.143.67.11
..
```

DNS time-to-live

need some way to remove out-of-date DNS records

DNS solution: time-to-live

number of seconds record is valid for

DNS exercise (1)

"www.cs.virginia.edu is 128.148.67.11 for next 86400 seconds"

(given record above) if sysadmin changes IP address DNS server returns for www.cs.virginia.edu, then what will happen to machines accessing website?

A. they'll start using the new address after 86400 seconds, and use the old one before then.

B. different machines will use the new address at different times, but no longer than 86400 seconds from when it changes

C. machines will start using the new address almost immediately, but after some small delay after it is changed

D. machines may keep using the old address until they are rebooted

E. something else?

DNS exercise (2)

if sysadmin wants to change the IP address of www.cs.virginia.edu, how do they do this without downtime?

they can change the IP address the server returns and/or the time-to-live?

what should they change and when to smoothly transition to a new address?

DNS exercise (3)

suppose initially

*.foo.com DNS server ('nameserver') = 10.2.3.4, valid 200 s

www.foo.com = 10.1.2.3, valid 100 s

if at time 0 seconds, changed to:

*.foo.com DNS server = 10.3.4.5, valid 100 s

www.foo.com DNS server = 10.3.5.1, valid 400 s

ex 0: when will new DNS server/www.foo.com start being used?

ex 1: when can we shut down old DNS server?

ex 2: when can we shut down old www.foo.com?

root servers (1)

(from IANA's website)

List of Root Servers

HOSTNAME	IP ADDRESSES	OPERATOR
a.root-servers.net	198.41.0.4, 2001:503:ba3e::2:30	Verisign, Inc.
b.root-servers.net	170.247.170.2, 2801:1b8:10::b	University of Southern California, Information Sciences Institute
c.root-servers.net	192.33.4.12, 2001:500:2::c	Cogent Communications
d.root-servers.net	199.7.91.13, 2001:500:2d::d	University of Maryland
e.root-servers.net	192.203.230.10, 2001:500:a8::e	NASA (Ames Research Center)
f.root-servers.net	192.5.5.241, 2001:500:2f::f	Internet Systems Consortium, Inc.
g.root-servers.net	192.112.36.4, 2001:500:12::d0d	US Department of Defense (NIC)
h.root-servers.net	198.97.190.53, 2001:500:1::53	US Army (Research Lab)
i.root-servers.net	192.36.148.17, 2001:7fe::53	Netnod
j.root-servers.net	192.58.128.30, 2001:503:c27::2:30	Verisign, Inc.
k.root-servers.net	193.0.14.129, 2001:7fd::1	RIPE NCC
l.root-servers.net	199.7.83.42, 2001:500:9f::42	ICANN
m.root-servers.net	202.12.27.33, 2001:dc3::35	WIDE Project

root servers (2)

(from root-servers.org)



anycast root servers

many root servers have multiple separate sites

...but one IPv4/IPv6 address

which one you get = which one routed to

multiple sites with BGP announcements for IP

- often with care taken to limit how far route goes

idea called 'anycast'

- get to 'any' of several servers

also used for some public recursive DNS servers

- 1.1.1.1 (CloudFlare), 8.8.8.8 (Google), 208.67.222.222 (Cisco), ...

DNS registries

IANA and affiliates publish root zone file

- same organization as for IP address, AS numbers
- (but domain names are lot more politically active)

For usable top-level domains, there are *registries*:

- VeriSign (VA company) for `COM`, `NET`
- Educause (non-profit) for `EDU`
- Registry Services, LLC (under contract with the US Dept of Commerce) for `US`
- ...

many registries allow for many *registrars* who can sell new domains

DNS resource records

resource record (RR) = single DNS database ‘entry’

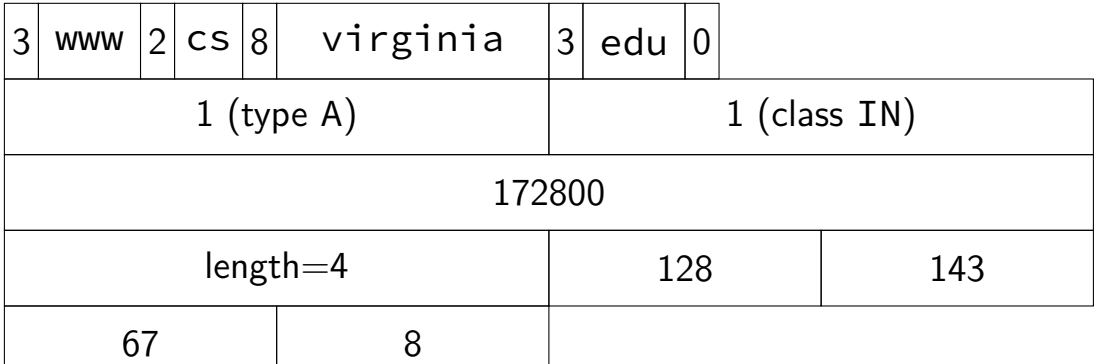
has standard text and binary representation

- text used for server config files and display
- binary representation used for network protocol

DNS record example (text)

`www.cs.virginia.edu.` `172800` `IN` `A` `128.143.67.8`

domain *name* sequence of *labels* empty label at end	time-to-live in seconds	*class* **IN**ternet	*type* v4 **A**ddr	*value*



DNS name format



zero or more: length of $K > 0$ followed by K character case-insensitive *label*

labels limited to 64 bytes

then either:

length of 0 to indicate end-of-name *or*
"pointer" to earlier name in message

pointer encoded as `0xc000` + byte number in message (big endian)
upper two bits being set prevents confusion with length

selected RR types

text	id	purpose	data format
A	1	IPv4 address	32-bit integer (big-endian)
AAAA	28	IPv6 address	128-bit integer (big-endian)
NS	2	authoritative name server	domain name
CNAME	5	‘canonical name’	domain name
TXT	16	text string	arbitrary string
SRV	33	service location	priority, weight, domain name, port

CNAME

CNAME = delegate name to another name

output of `dig canvas.its.virginia.edu a`:

```
canvas.its.virginia.edu. 2711   IN      CNAME
    universityofvirginia-vanity.instructure.com.
```

canvas.its.virginia.edu = universityofvirginia.vanity.instructure.com

```
universityofvirginia-vanity.instructure.com. 238 IN CNAME \
    canvas-pdx-prod-c354-1908777142.us-west-2.elb.amazonaws.com.
```

universityofvirginia.vanity.instructure.com = canvas-pdx-prod-c354-1908777142.us-west-2.elb.amazonaws.com.

```
canvas-pdx-prod-c354-1908777142.us-west-2.elb.amazonaws.com. 2 IN A 34.208.211.157
canvas-pdx-prod-c354-1908777142.us-west-2.elb.amazonaws.com. 2 IN A 44.238.73.244
canvas-pdx-prod-c354-1908777142.us-west-2.elb.amazonaws.com. 2 IN A 52.26.9.245
```

NS

```
virginia.edu. 3600    IN      NS      uvaarpa.virginia.edu.
virginia.edu. 3600    IN      NS      nom.virginia.edu.
virginia.edu. 3600    IN      NS      eip-01-aws.net.virginia
```

these three servers are authoritative for virginia.edu

(but need their A or AAAA records to actually contact them)

SRV

```
_ldap._tcp.virginia.edu. 600    IN      SRV     0 100 389 vadc
_ldap._tcp.virginia.edu. 600    IN      SRV     0 100 389 vadc
```

virginia.edu's LDAP servers are vadcv4.virginia.edu port 389 and vadc5.virginia.edu port 389

(LDAP = Lightweight Directory Access Protocol)

example: lookup email ID by name

SRV records were late...

SRV records seem like something we'd use all the time...

but only used with a few protocols

why?

SRV records added late (1996–2000)

sending email already had its own record type (MX)

poor support for querying them in standard networking libraries

hard to get access to add them in many organizations

dig virginia.edu txt

```
virginia.edu.           3548    IN      TXT     "google-site-verification=zEwuk4FIG8_vtv2BZJ
virginia.edu.           3548    IN      TXT     "cisco-ci-domain-verification=268b991de35894
virginia.edu.           3548    IN      TXT     "miro-verification=9f3238c881466b3ccb99c4347
virginia.edu.           3548    IN      TXT     "MS=ms40126609"
virginia.edu.           3548    IN      TXT     "onetrust-domain-verification=d901adf0911647
virginia.edu.           3548    IN      TXT     "google-site-verification=rePHmrM7FEfSw62DHq
virginia.edu.           3548    IN      TXT     "v=spf1 a mx ip4:52.254.56.82 ip4:52.137.91.
virginia.edu.           3548    IN      TXT     "docusign=6547d080-1c40-427c-8736-fafa466ff7
virginia.edu.           3548    IN      TXT     "apple-domain-verification=zllSamqSFAw4lNQo"
virginia.edu.           3548    IN      TXT     "v=msv1 t=537499f256ea46bd386f7543d18d28"
virginia.edu.           3548    IN      TXT     "atlassian-domain-verification=zeQ45p2Vl1fXe
virginia.edu.           3548    IN      TXT     "00256316"
virginia.edu.           3548    IN      TXT     "yYMwnn+VY4w6aVf+cE4888ppz+MDXBKnaI0m5eteiMG
virginia.edu.           3548    IN      TXT     "apple-domain-verification=mROwKUEMfwLBfltC"
virginia.edu.           3548    IN      TXT     "e2ma-verification=rx8eb"
virginia.edu.           3548    IN      TXT     "ZOOM_verify_PuU-eQzyQ7ONrij_vTP72A"
virginia.edu.           3548    IN      TXT     "sending_domain1023271=f4a69653389ff4543a64e
```

TXT record usage

TXT records ‘just’ hold arbitrary strings

lots of machine-based usage of TXT records

probably in part because it’s too slow/hard to add new record types

TXT example: SPF

```
virginia.edu. 3548    IN      TXT "v=spf1 a mx
    ip4:52.254.56.82 ip4:52.137.91.139
    ip4:128.143.125.90 ip4:128.143.125.91
    include:spf.protection.outlook.com
    include:_spf.google.com
    include:spf.elluciancloud.com ~all"
```

SPF protocol for specifying who can send email from domain

why in TXT record instead of its own type?

RFC 6686, Appendix A excerpt

At the time of SPF's initial development, the prospect of getting an RRTYPE allocated for SPF was not seriously considered, partly because doing so had high barriers to entry....

Later, after RRTYPE 99 was assigned ...a plan was put into place to effect a gradual transition to using RRTYPE 99 instead of using RRTYPE 16. This plan failed to take effect for four primary reasons:...

1....existing nameservers (and, in fact, DNS-aware firewalls) would drop or reject requests for unknown RRTYPEs ...

2. many DNS provisioning tools ...were, and still are, typically lethargic about adding support for new RRTYPEs

...

primary and secondary servers

usually have multiple DNS servers for each zone

- needed to handle outages

DNS protocol supports *zone transfers* to synchronize them

- DNS query for type AXFR returns all RRs as special case
- (most public DNS servers block this from 'normal'users)

one server designed as 'primary'; others are 'secondary'

special SOA (start of authority) record provides metadata about zone

- name of primary server to contact
- serial number for current version (to quickly check if update)
- how long to wait between updates, etc.

DNS messages (1)

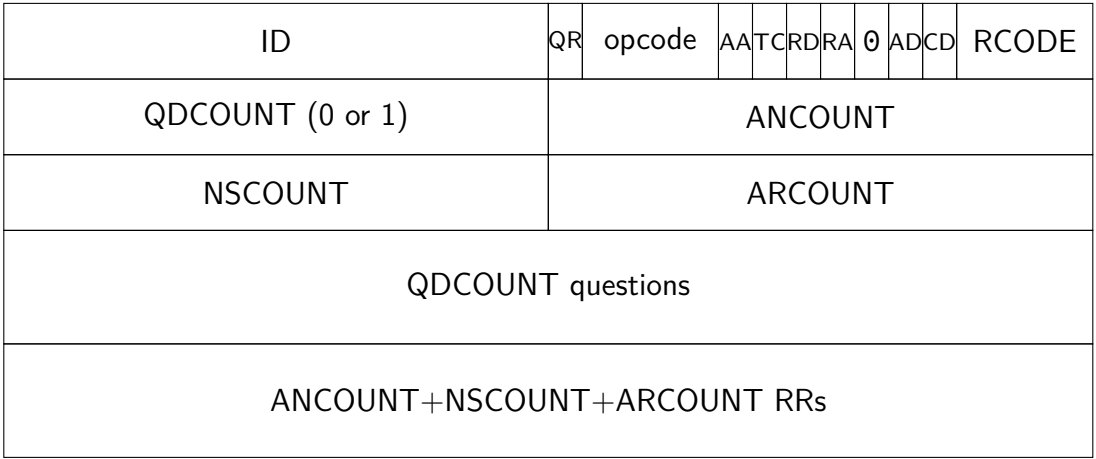
client sends query to server

server responds with response

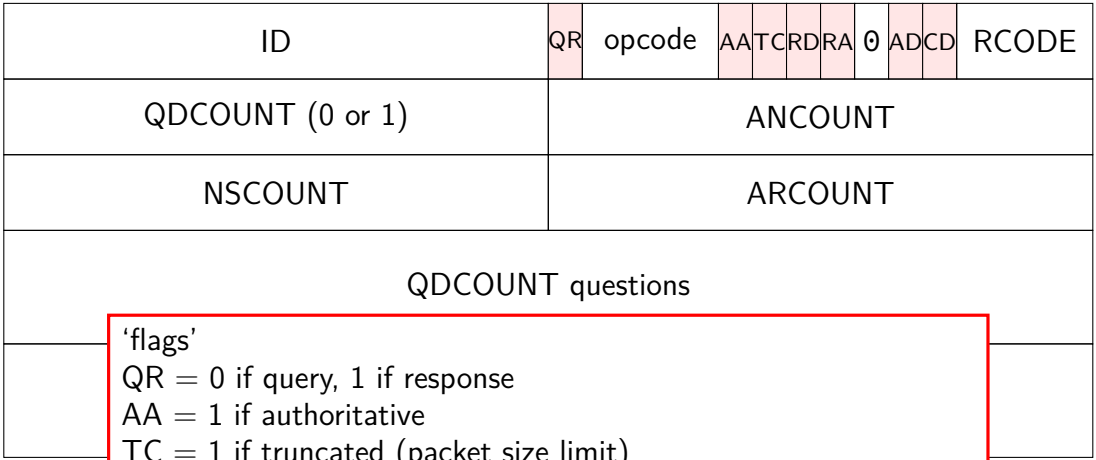
response and query have same format

- but different fields set/used

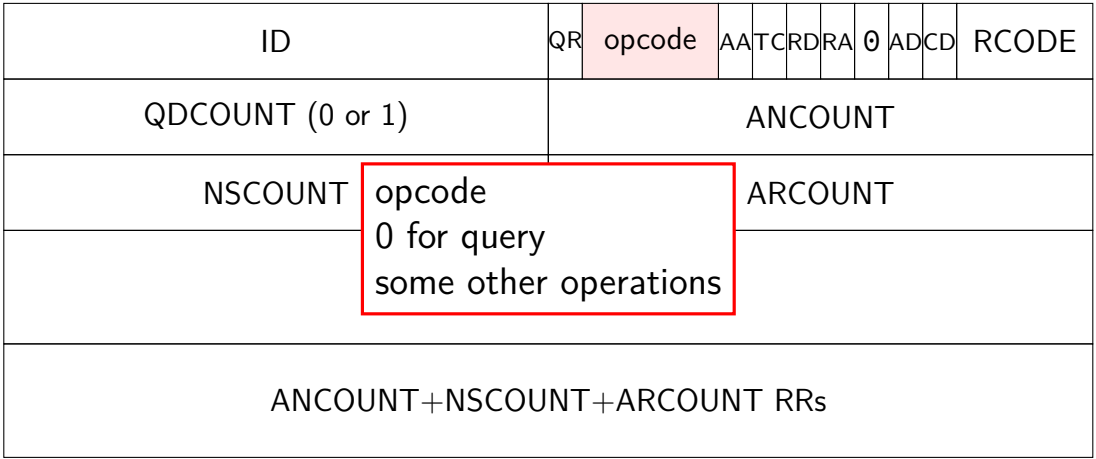
DNS header


ID
QR
opcode
AA
TC
RD
RA
0
AD
CD
RCODE
QDCOUNT (0 or 1)
ANCOUNT
NSCOUNT
ARCOUNT
QDCOUNT questions
ANCOUNT+NSCOUNT+ARCOUNT RRs

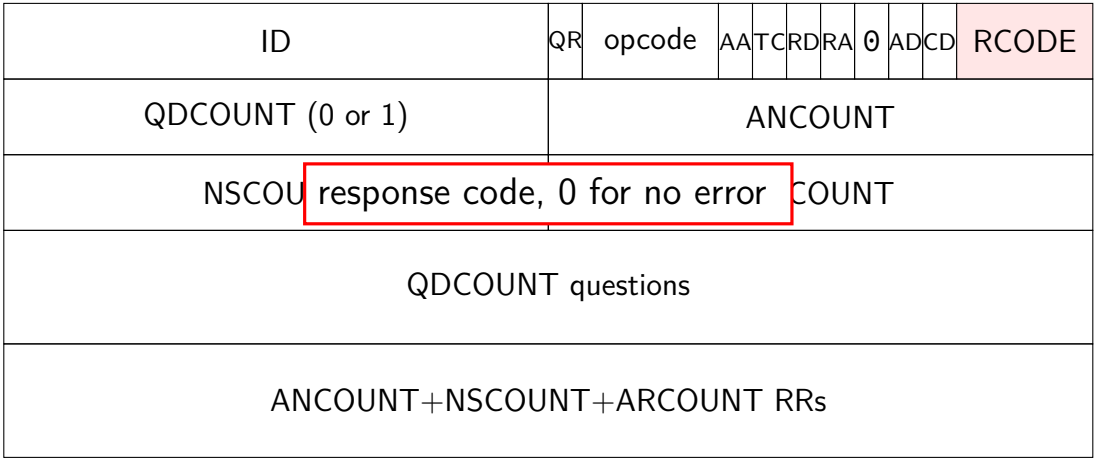
DNS header


ID
QR
opcode
AA
TC
RD
RA
0
AD
CD
RCODE
QDCOUNT (0 or 1)
ANCOUNT
NSCOUNT
ARCOUNT
QDCOUNT questions
'flags'
QR = 0 if query, 1 if response
AA = 1 if authoritative
TC = 1 if truncated (packet size limit)
RD = 1 if query wants recursion (contact other servers)
RA = 1 if server supports recursion

DNS header


ID
QR
opcode
AA
TC
RD
RA
0
AD
CD
RCODE
QDCOUNT (0 or 1)
ANCOUNT
NSCOUNT
ARCOUNT
opcode
0 for query
some other operations
ANCOUNT+NSCOUNT+ARCOUNT RRs

DNS header


ID
QR
opcode
AA
TC
RD
RA
0
AD
CD
RCODE
QDCOUNT (0 or 1)
ANCOUNT
NSCOU
response code, 0 for no error
COUNT
QDCOUNT questions
ANCOUNT+NSCOUNT+ARCOUNT RRs

caching ‘domain name does not exist’

when domain does not exist (RCODE=3, “NXDOMAIN”):

(RFC 2308)

response should include SOA record

- recall: indicates primary server for zone and how often secondaries should update

can cache NXDOMAIN based on minimum of TTL and SOA record’s update frequency

DNS message

header

0 or 1 question

- domain name + class + type
- (RR without TTL or value)
- special type values for ANY, some other things

ANCOUNT RRs in ‘answer section’:

- CNAMEs (even if that’s not the question type)
- matches for domain name (or CNAME) + class + type

NSCOUNT+ARCOUNT RRs in ‘authority section’ and ‘additional section

- not always included; usually NS or SOA records plus corresponding A/AAAA recods

DNS over ...

UDP port 53

- send DNS message
- reply with DNS message back

TCP port 53

- send length (2B, big endian) + message
- reply with length (2B, big endian) + message
- (+ repeat)

over HTTPS (RFC 8484)

- using `https://server/dns-query`

DNS UDP size limit

```
$ dig virginia.edu txt +notcp +all
;; Truncated, retrying in TCP mode.

; <<>> DiG 9.18.28-0ubuntu0.22.04.1-Ubuntu <<>> virginia.edu txt +notcp +all
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 53043
;; flags: qr rd ra; QUERY: 1, ANSWER: 17, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;virginia.edu.                          IN        TXT

;; ANSWER SECTION:
virginia.edu.                3537        IN        TXT        "cisco-ci-domain-ver
....
```

DNS UDP size limit

512 byte size limit for UDP

- if exceeded, set TC (truncate) flag and truncate response
- clients supposed to retry with TCP

extensions to DNS increase limit

- (but have to be opted into by clients)

explains why only 13 root server names

- 13 NS RRs + 13 A RRs $\approx$ 480 bytes (w/ name compression)

eDNS(0) (RFC 6891)

extend DNS protocol by sending psuedo-RR

- added in additoinal section of requests
- name `.` (empty)
- type OPT/44
- CLASS, TTL fields reused for other stuff

specifies:

- max size over UDP
- extra bits for RCODE
- additoinal, variable length data

DNSSEC

public key infrastructure for DNS

single set of root keys from ICANN/IANA

- no certificate authorities like web PKI

digital signature for each delegation to new servers

- delegation to new zone includes keys for that zone
- chain of signatures DNS client can check
- everything can still be cached

makes DNS messages a lot bigger

DNSSEC and missing records

tricky problem: validating ‘not present’ responses

DNSSEC has multiple options:

- signed ‘no result between W.Y.Z, type Q and Z.Y.Z, type A’ message
- signed ‘no result with hash(?) = A < hash(X) < B = hash(?)’ message

can be generated in advance (with signing keys kept ‘offline’)

- can also generate dynamically to reveal less information

DNSSEC deployment: validation

queries supporting validation: approx. 35%

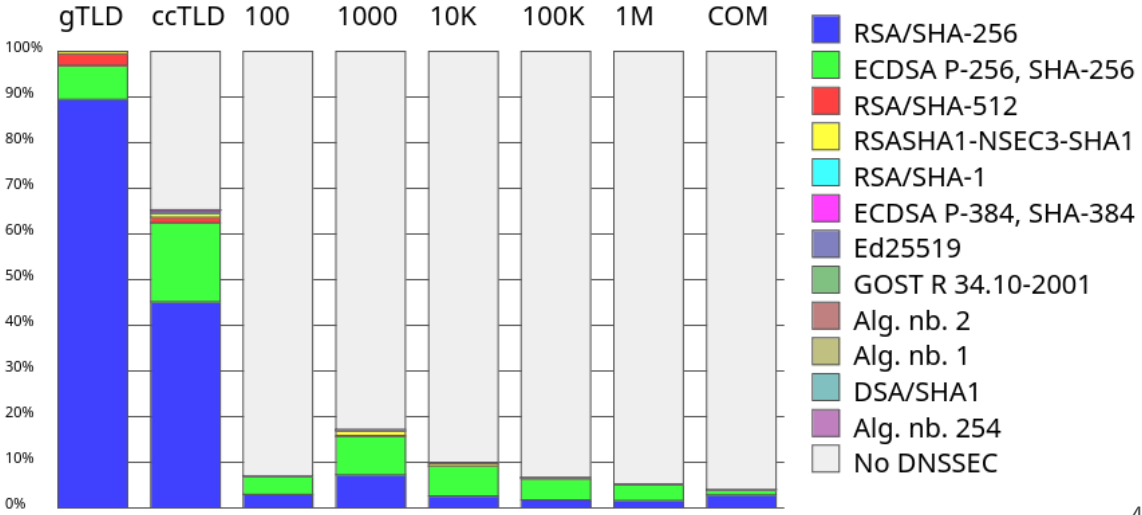
from `https://stats.labs.apnic.net/dnssec/`

approx. 45% recursive resolvers support

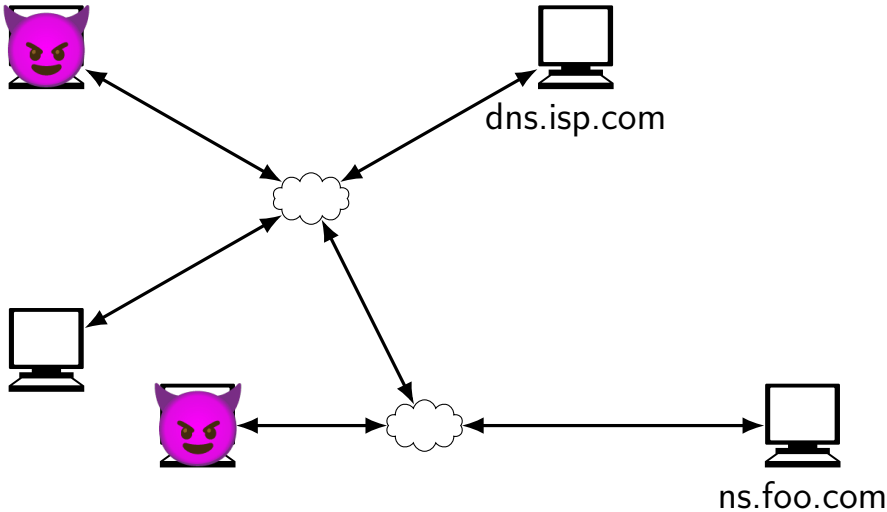
from `https://ithi.research.icann.org/`

DNSSEC deployment: signing

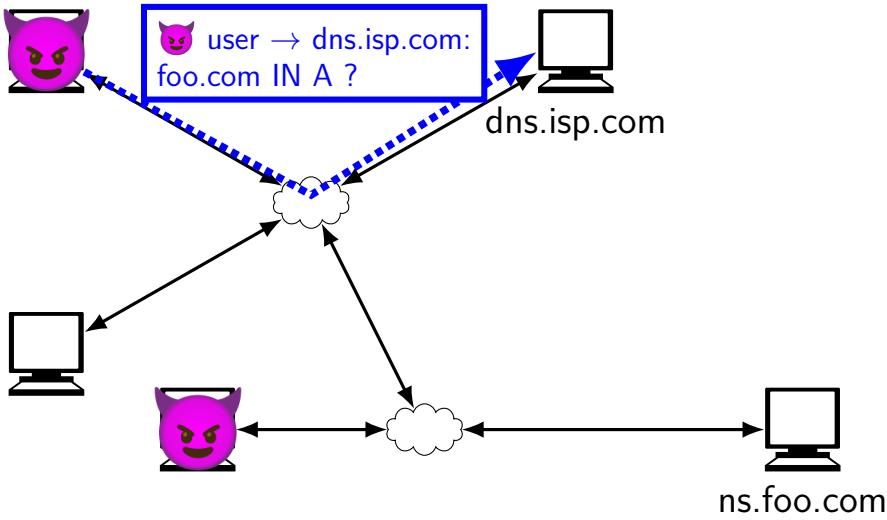
via `https://ithi.research.icann.org/graph-m11.html`:



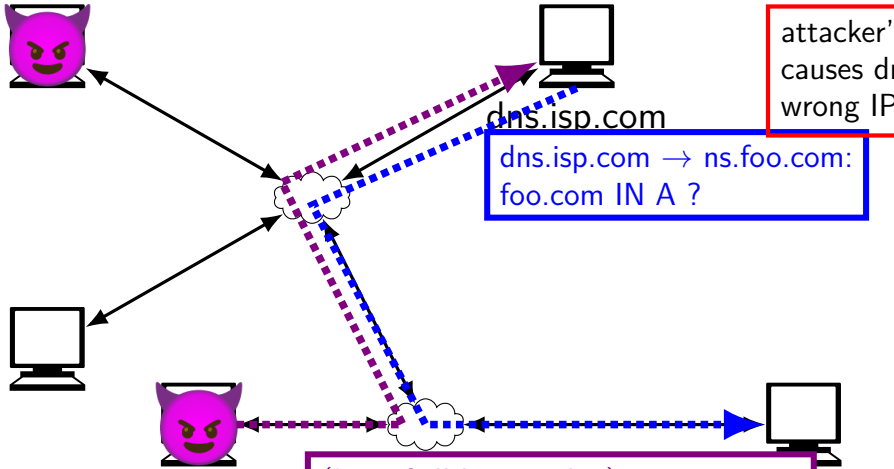
DNS cache poisoning



DNS cache poisoning

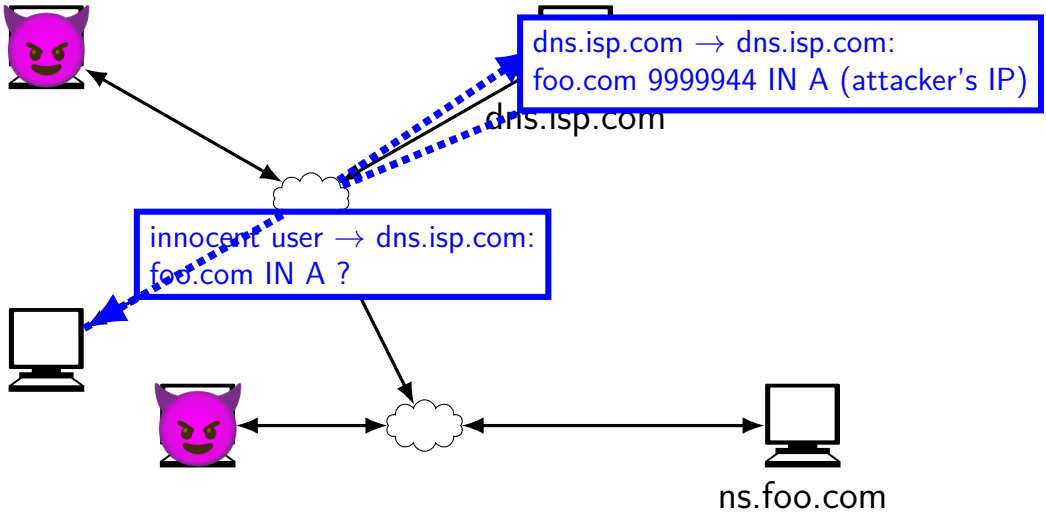


DNS cache poisoning



attacker’s ‘spoofed’ response causes dns.isp.com to record wrong IP

DNS cache poisoning



mitigating cache poisoning attacks

filter out packets with source address for where they come from?

- not feasible if real/spoofed packets forwarde through many other ISPs

use random port number for queries

- attacker can spoof many port numbers at once
- attacker can keep trying until they guess right

use random ID number in DNS query

- not good enough alone — attacker can guess often enough
- probably enough with random port?

add additional randomness to DNS query

- randomize capitalization (assuming it's returned the same in response)
- 'DNS cookie' extension (RFC 7873; assuming remote server supports it)

use TCP with random sequence number (slow)

reverse DNS (IPv4)

what's a domain name for IP 128.143.107.101?

special domain name: `101.107.143.128.in-addr.arpa`

...and PTR record type for this:

```
$ dig -x 128.143.107.101
...
101.107.143.128.in-addr.arpa. 3516 IN   PTR     eip-04-udc.net.virginia.edu.
...
$ dig eip-04-udc.net.virginia.edu a
...
eip-04-udc.net.virginia.edu. 3600 IN    A       128.143.107.101
...
```

might not be only name:

```
$ dig nom.virginia.edu a
...
nom.virginia.edu.       86400   IN      A       128.143.107.101
...
```

reverse DNS (IPv6)

```
$ dig -x 2607:f8b0:4004:c1d::65
...
5.6.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.d.1.c.0.4.0.0.4.0.b.8.f.7.0.6.2.ip6.arpa.
    3345 IN PTR ww-in-f101.1e100.net.
...
```

domain name system blocklists

historically common non-domain-name use of DNS

- I'm including to illustrate idea of 'other' DNS usage

database identifying (typically) IP addresses for filtering

used for spam/bot detection/prevention

- these days tend to expect payment for serious use

typical idea similar to reverse IP lookups

- use specific IP address to mark 'in' or 'not in' list

domain name system blocklists

example: Team Cymru's Bogon list identifies invalid addresses

```
# 248.1.1.1 is on blocklist
$ dig 1.1.1.248.v4.fullbogons.cymru.com a
...
1.1.1.248.bogons.cymru.com. 21600 IN    A       127.0.0.2
...

# 128.143.67.31 is not on blocklist
$ dig 31.67.143.128.v4.fullbogons.cymru.com a
...
;; ->>HEADER<<- opcode: QUERY, status: NXDOMAIN, id: 6153
...
```

internationalized domain names

`https://日本レジストリサービス.jp/`

how does this work?

becomes:
`https://xn--vckfdb7e3c7hma3m9657c16c.jp/`

encoding scheme called *punycode*

IDN homograph attacks

bankofamerica.com

xn--bnkofamerica-x9j.com

a = U+0430 = CYRLLIC SMALL LETTER A

defenses against homograph attacks

at registries, restrict domain registration

- disallow mixed scripts (e.g. latin and cyllric)
- test if looks identical to registered domains

at browsers, restrict display in non-`xn--...` form

- allow-list for 'good' top-level domains (e.g. `.gr`, `.jp`, etc.)
- otherwise, only allow known non-confusing combinations

backup slides